

ARM Exam Quick Reference

Searchable exam, solution-pattern, adaptation, and verification reference

AAPCS

R0-R3 arguments and scratch; R0 result; preserve R4-R11; save LR before nested BL; restore SP exactly; keep public call boundaries eight-byte aligned.

Addressing

byte index x1; halfword index x2; word index x4; row-major matrix index = row * columns + column.

Signedness

Use LDRSB/LDRSH and signed conditions for signed values. Use LDRB/LDRH and unsigned conditions for unsigned values.

Interrupts

Acknowledge the source, capture bounded state, publish a volatile event, and do longer work in foreground unless the paper says otherwise.

Minimal-change checklist

Change configuration first; keep the ABI; localize encoding/recurrence rules; rerun guards, boundaries, ABI checks, state timelines, simulator execution, and the hardware build.

Lookup

abi:nonleaf: PAT-AAPCS-NONLEAF-001
risk:stack-alignment: PAT-AAPCS-STACK-SAFETY-001
mem:matrix-row-major: PAT-MEM-MATRIX-ROW-MAJOR-001
flow:nested-loop: PAT-FLOW-NESTED-LOOP-001
risk:irq-shared-state: PAT-STATE-IRQ-HANDOFF-001
board:gpio: PAT-GPIO-EVENT-001
alg:recurrence: PAT-ALG-RECURRENCE-001
board:timer: PAT-TIMER-OWNERSHIP-001
timing:periodic: PAT-TIMER-PERIODIC-001
state:event-loop: PAT-STATE-EVENT-LOOP-001
cpu:flags: PAT-CPU-FLAGS-001
mem:word-array: PAT-MEM-WORD-ARRAY-001
mem:byte-array: PAT-MEM-BYTE-ARRAY-001
state:debounce: PAT-STATE-DEBOUNCE-001
board:adc: PAT-ADC-SAMPLE-001
board:dac: PAT-DAC-STREAM-001
timing:free-running: PAT-TIMER-FREE-RUNNING-001
alg:graph-search: PAT-ALG-GRAPH-SEARCH-001
alg:frequency-count: PAT-ALG-FREQUENCY-COUNT-001
flow:early-break: PAT-FLOW-EARLY-BREAK-001
abi:stacked-args: PAT-AAPCS-STACKED-ARGS-001
abi:four-register-args: PAT-AAPCS-FOUR-ARGS-001
alg:fixed-point: PAT-ALG-FIXED-POINT-001

board:joystick: PAT-GPIO-JOYSTICK-001

alg:sorting: PAT-ALG-SORTING-001

cpu:svc: PAT-CPU-SVC-001

risk:vector-ownership: PAT-TIMER-VECTOR-OWNERSHIP-001

cpu:exception-frame: PAT-CPU-EXCEPTION-FRAME-001